

Orthodontic Considerations, Space Maintainers

Ian Marion, DDS

Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Objectives

- List methods of assessing occlusion in children
- Describe the transition from primary to permanent dentition
- Identify problems detectable in the primary and mixed dentitions
- Describe different space maintenance options
- Be able to select space maintainers based on missing teeth
- List different interceptive orthodontic treatment options

Outline

Occlusal Assessment

Physiologic Eruption

Space Concerns

Space Interventions

Early Orthodontic Concerns

Early Orthodontic Interventions

Early Orthodontic Eval

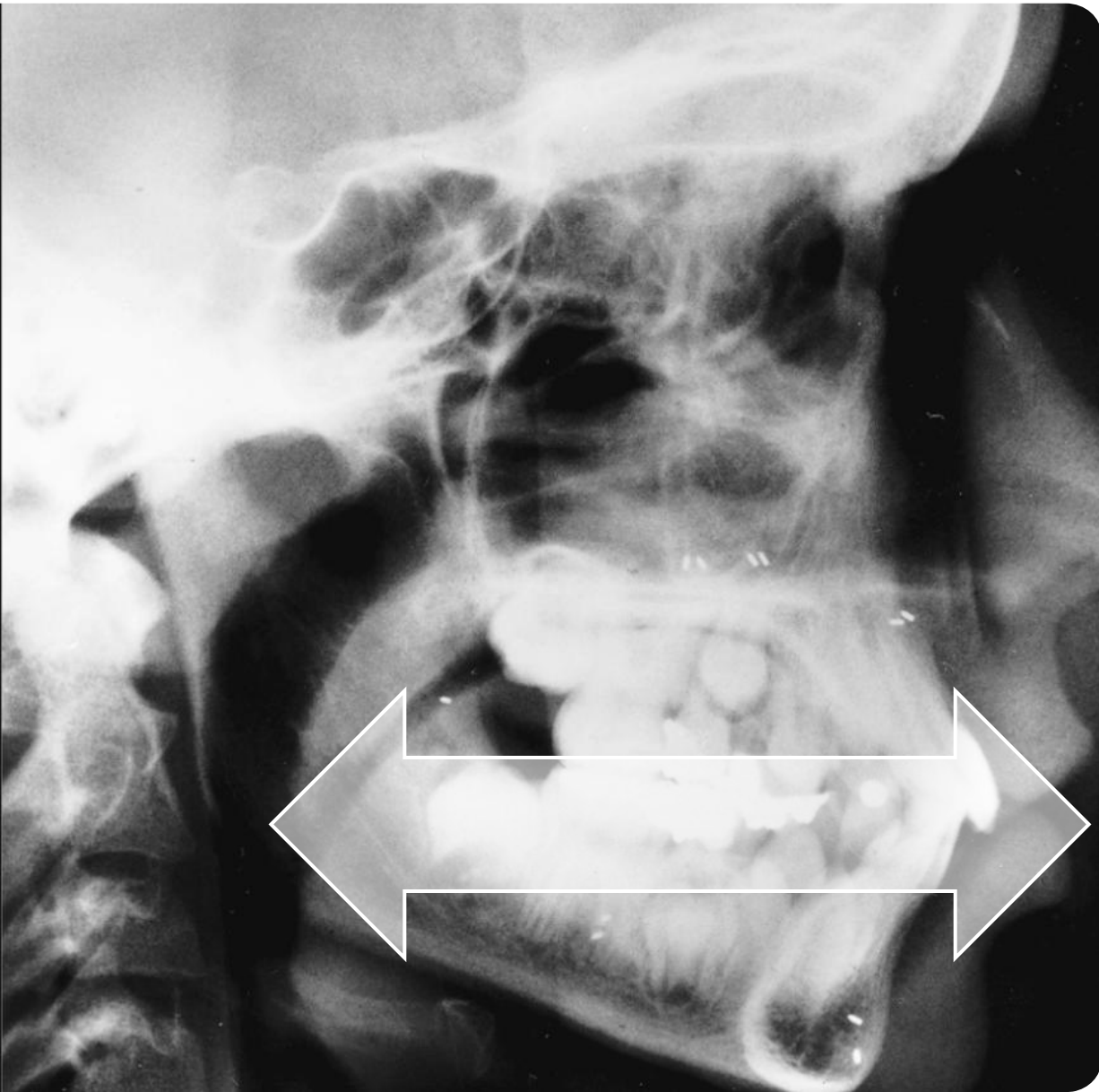
Occlusion in Children

We look at
a patient in
3
dimensions

Anteroposterior

Transverse

Vertical



Anteroposterior

Anteroposterior

- Positioning of the dentition in an AP direction, particularly between the arches is a key assessment
- Both anterior and posterior teeth are involved
- Molar class
- Overjet
- Anterior Crossbites

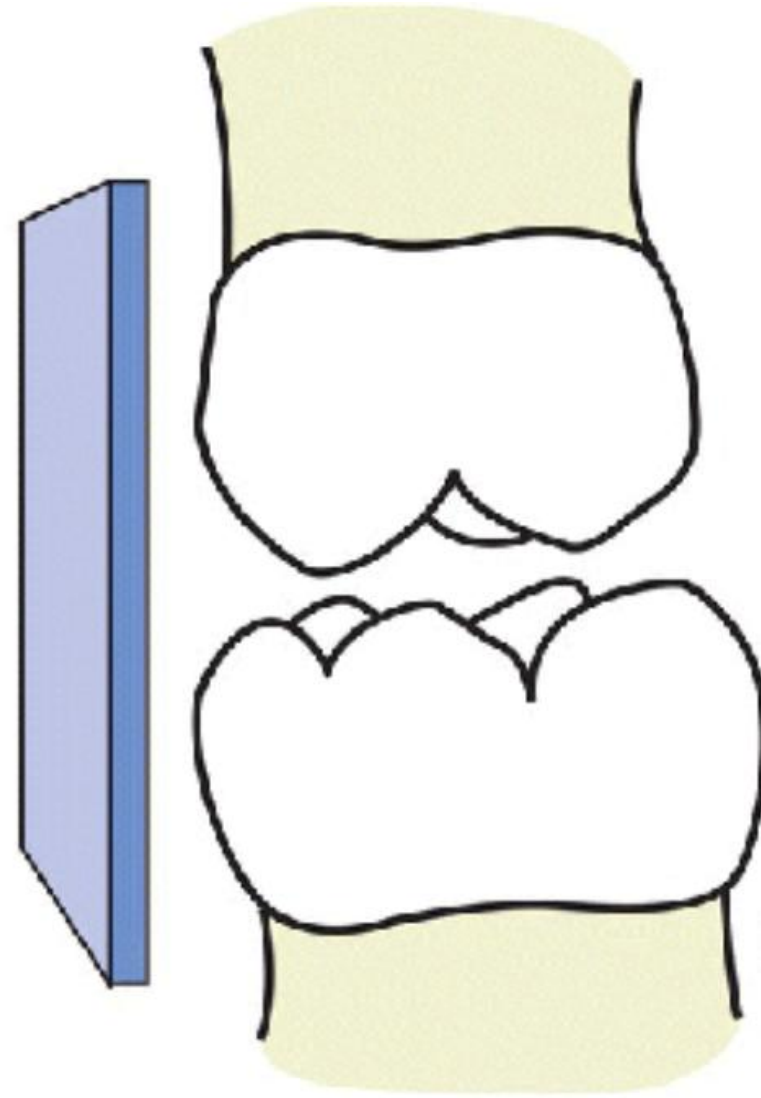
Primary Molar Classification

- Primary Molars are assessed based on the position of the mandibular distal terminal plane compared to the maxillary
- The direction we “step down” to

Flush Terminal Plane

- Distal plane of the lower second molar is in line with the distal plane of the upper second molar
- This is the most common primary occlusion

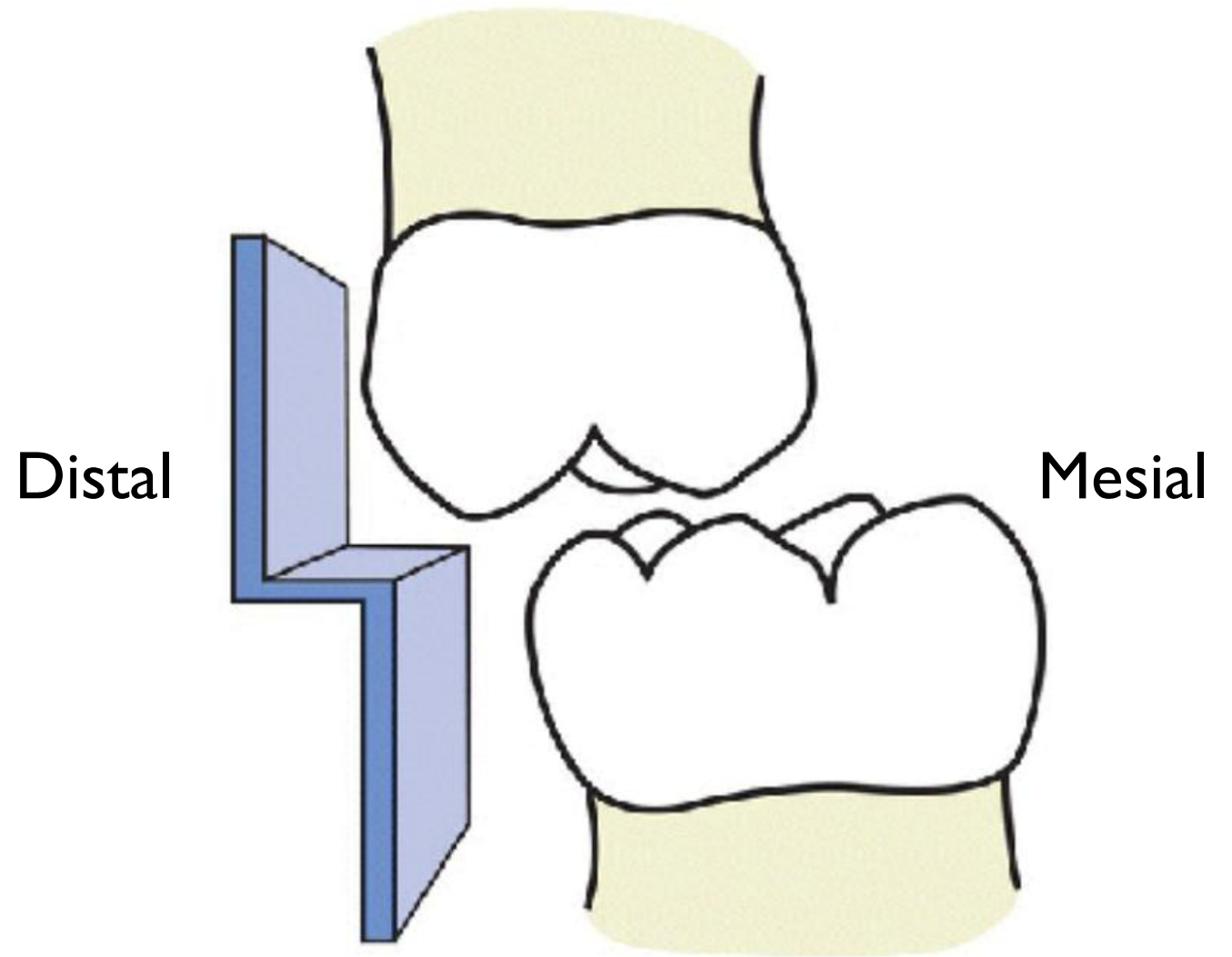
Distal



Mesial

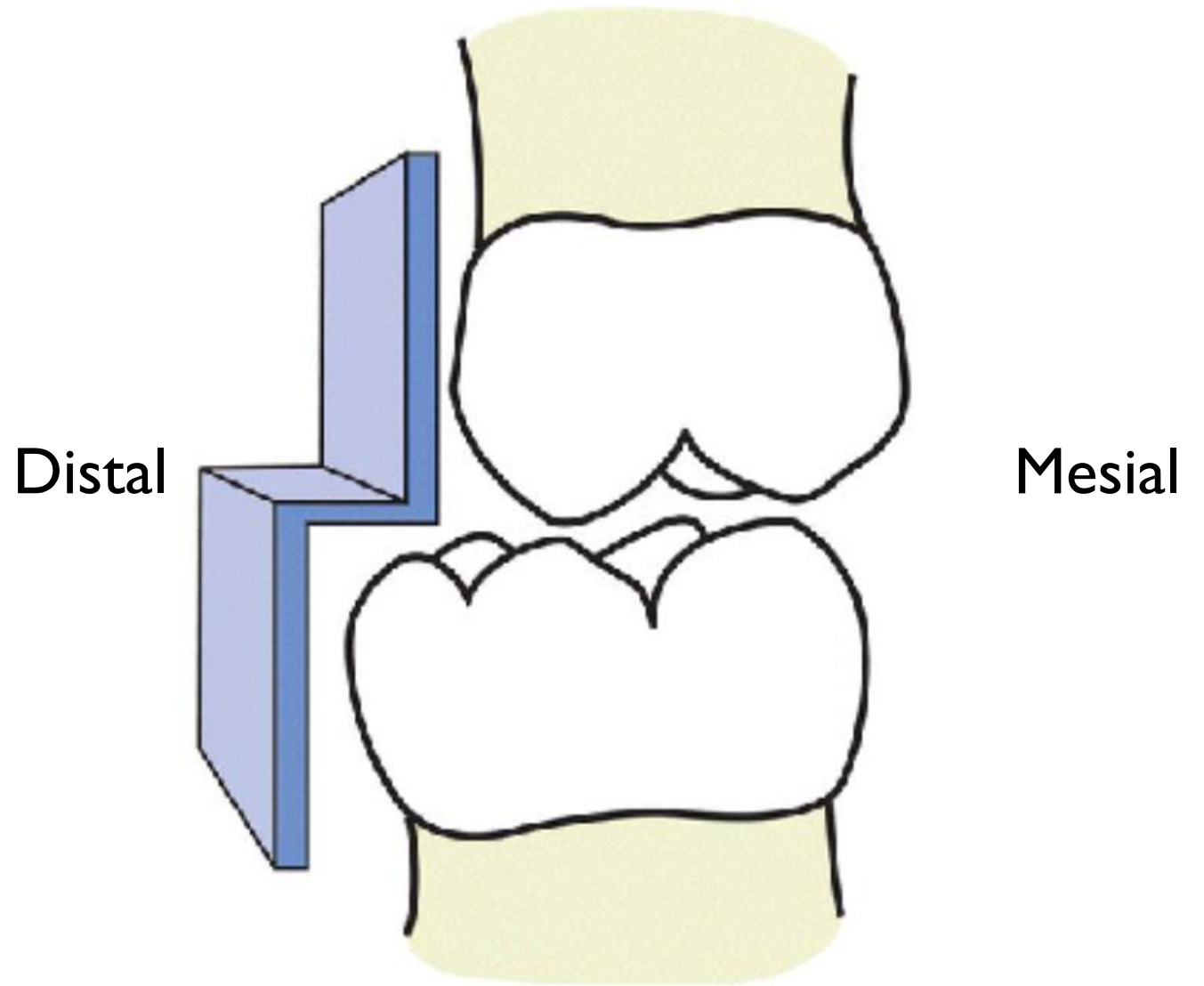
Mesial Step

- Distal plane of the lower second molar is mesial to the distal plane of the upper second molar
- This is the second most common primary occlusion



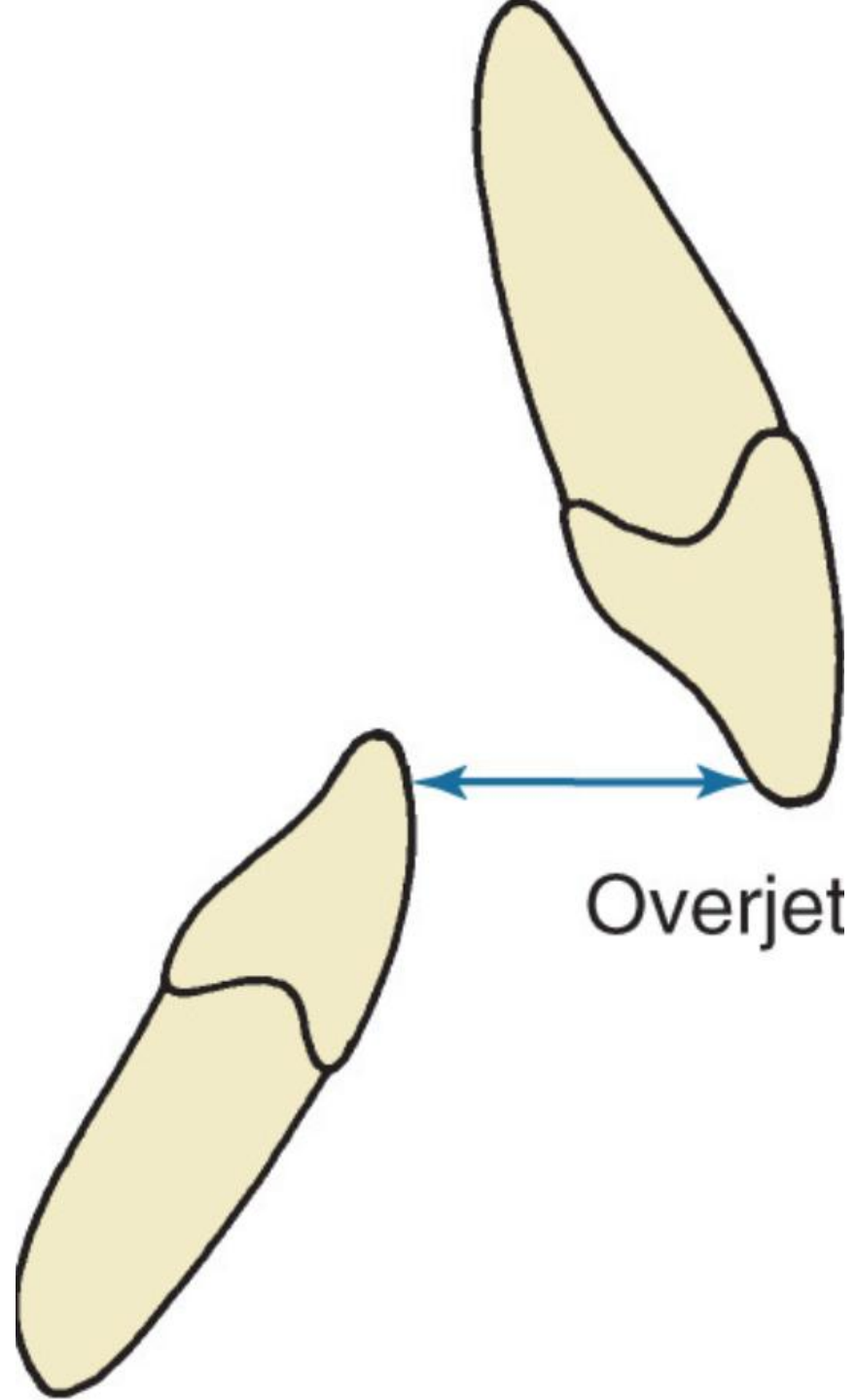
Distal Step

- Distal plane of the lower second molar is distal to the distal plane of the upper second molar
- This is the least common primary occlusion



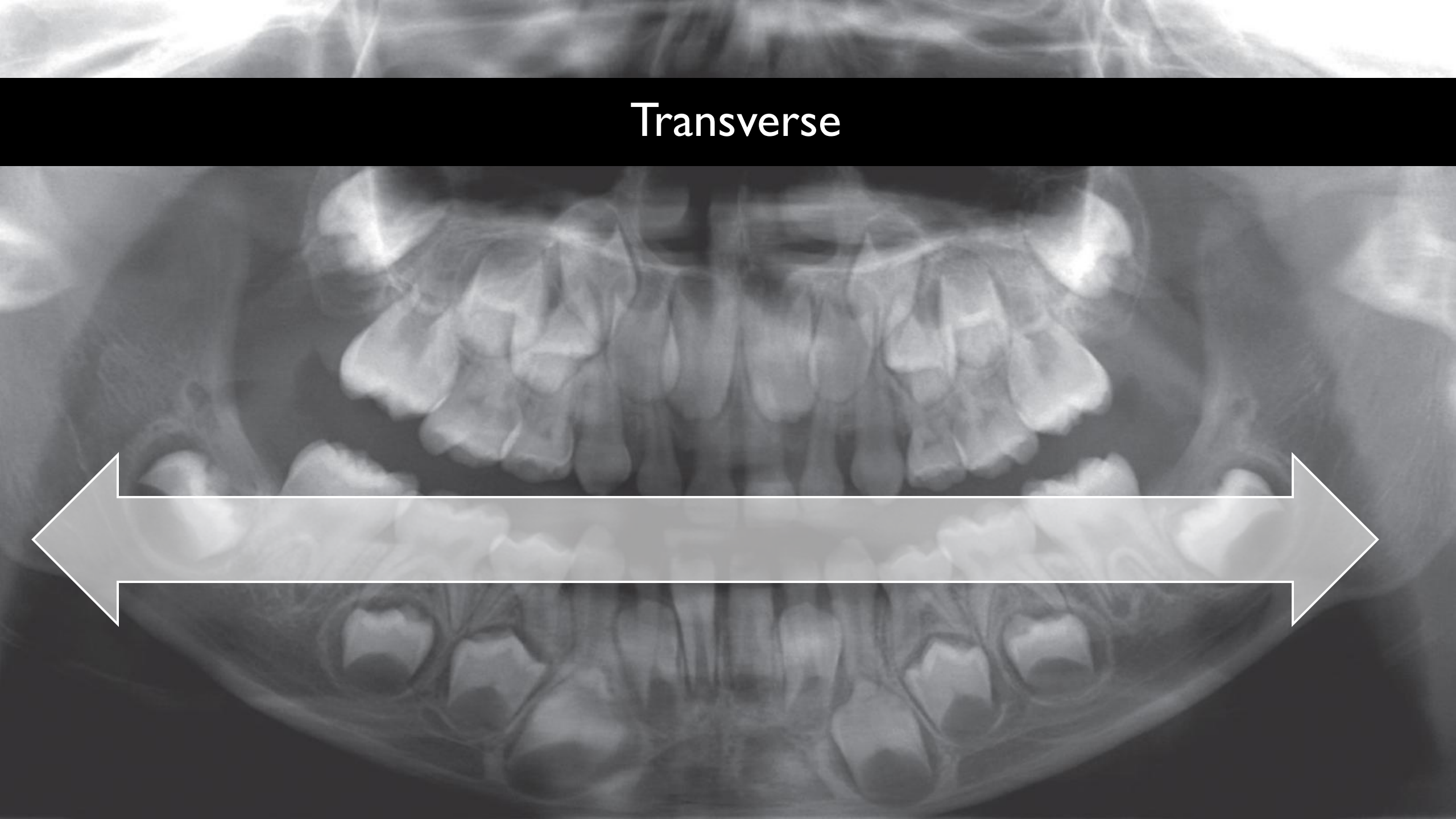
Overjet

- Overjet is a measurement of the span between the upper incisors and the lower in the anterior to posterior direction
- Can measured in mm
- Valuable to note if overjet is negative, 0, positive, or excessive





Transverse

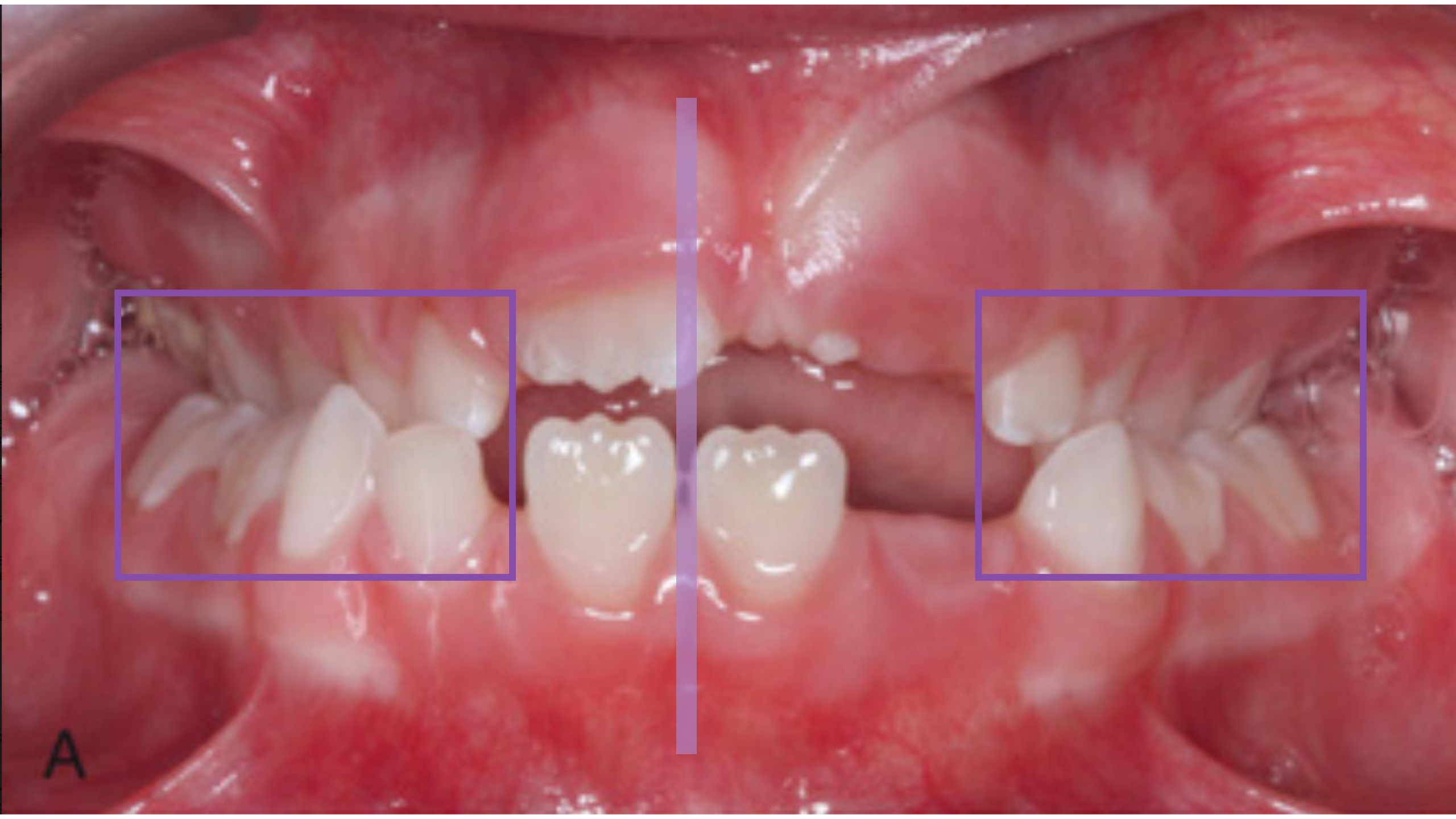


Transverse

- Assessment of arch width and the relative positions of the maxilla and mandible in a lateral direction
- Midline evaluation
- Posterior Crossbites

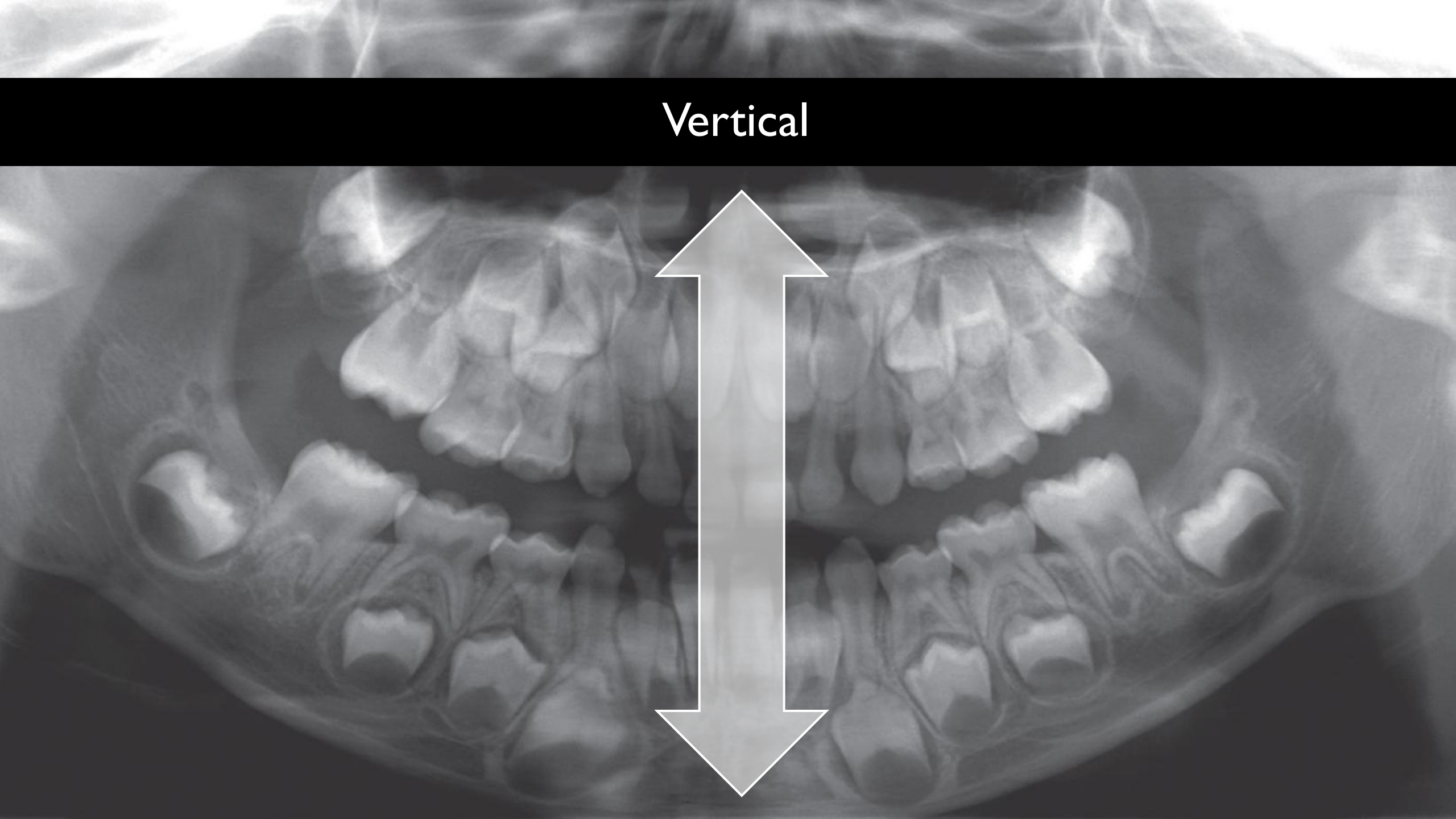


A



A

Vertical

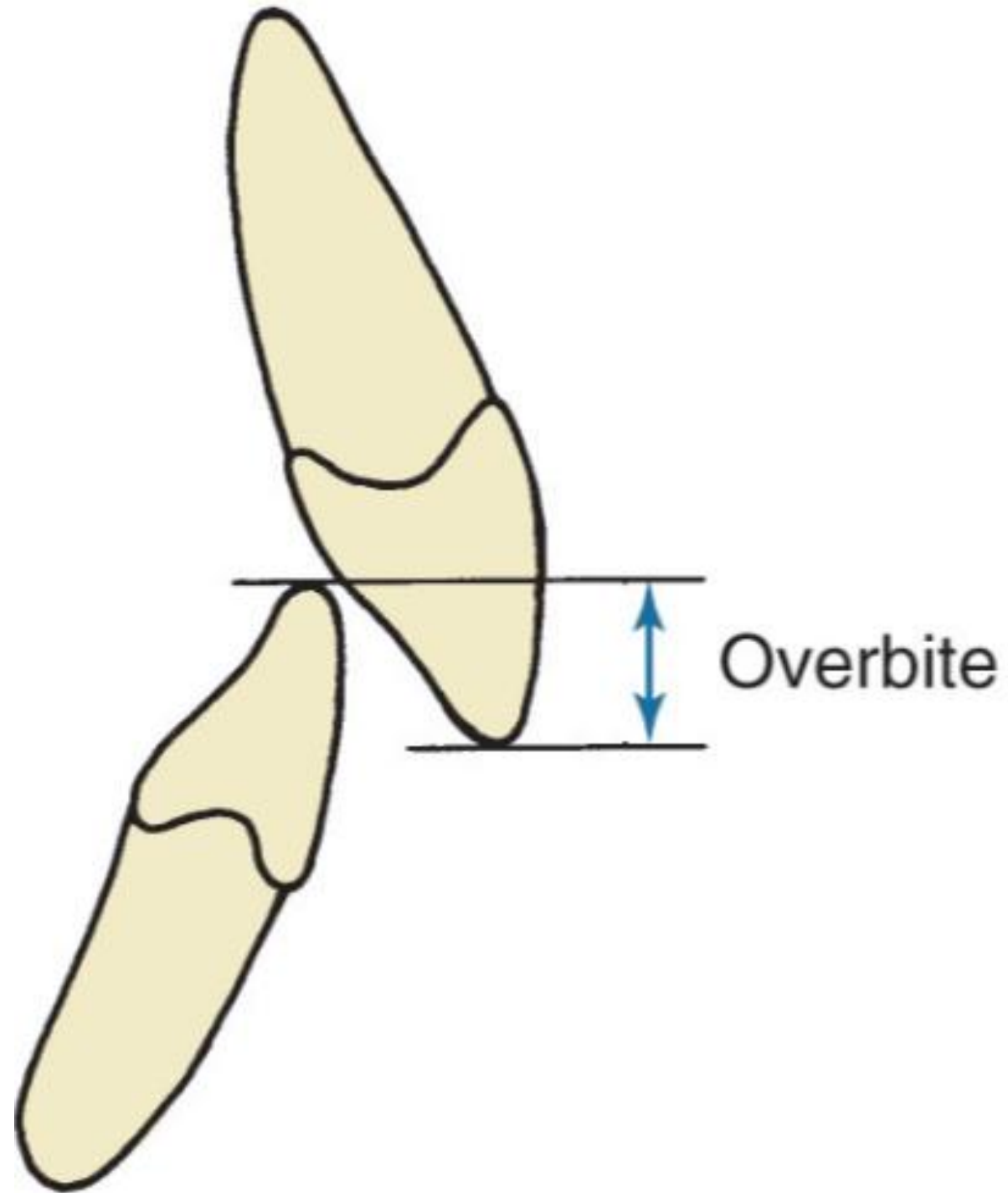


Vertical

- Assessment of the relative positions of the maxilla and mandible in an inferior to superior direction
- Overbite
- Open bites

Overbite

- Overbite is a measurement of the span between the incisal edges of the upper incisors and the lower in the inferior to superior direction
- Can measured in mm or %
- Again valuable to note if overbite is negative, 0, positive, or excessive





Spacing Assessment

A comprehensive spacing eval involves clinical and radiographic assessment in multiple dimensions

Baume Type

Primate Space

Leeway Space

“E” Space

Baume Type

Type I “Spaced” occurs
in $2/3^{\text{rd}}$ of primary
dentitions

Type II “Closed” occurs
in the remaining $1/3^{\text{rd}}$

Type 1 Spaced

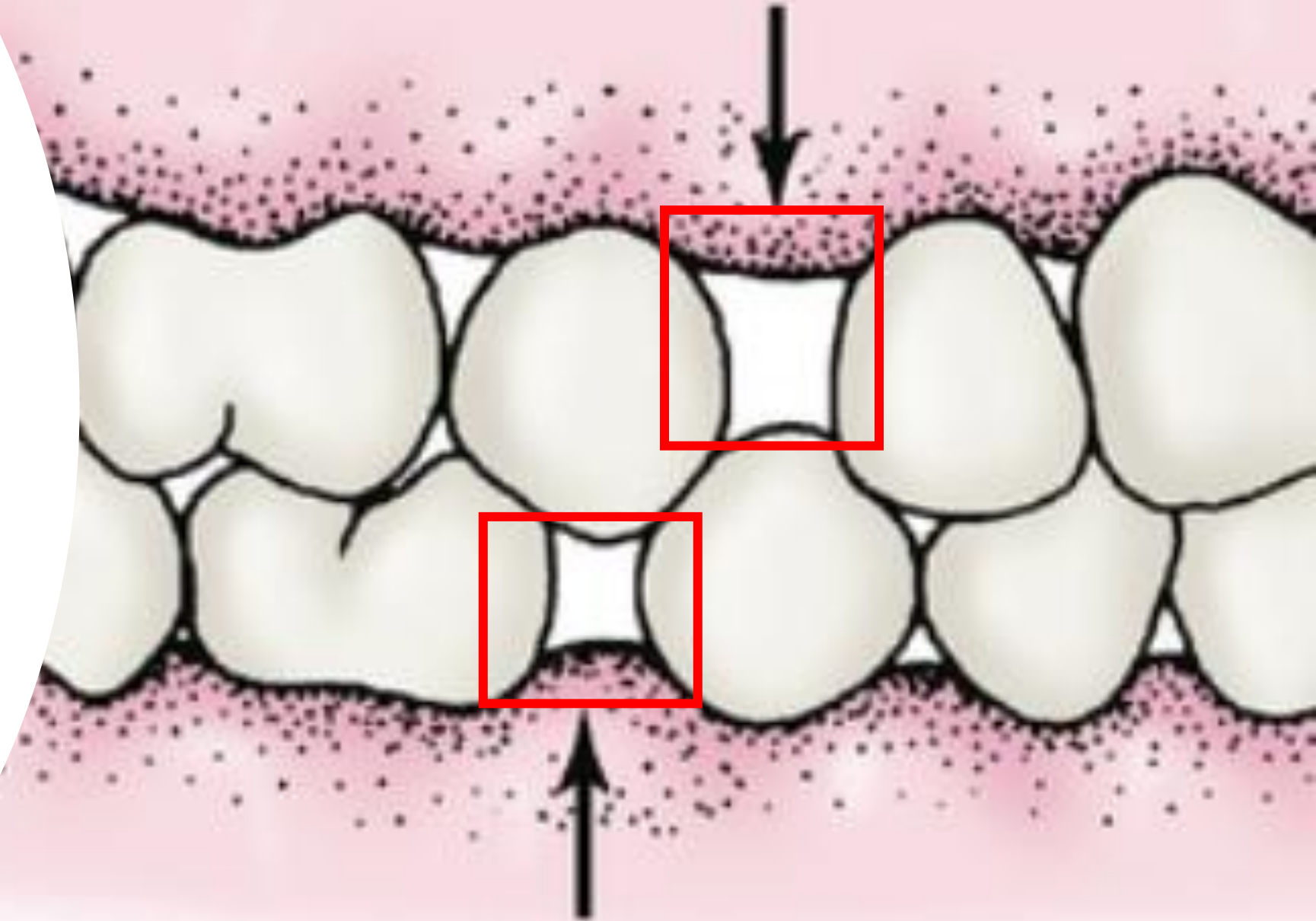


Type 2 Closed



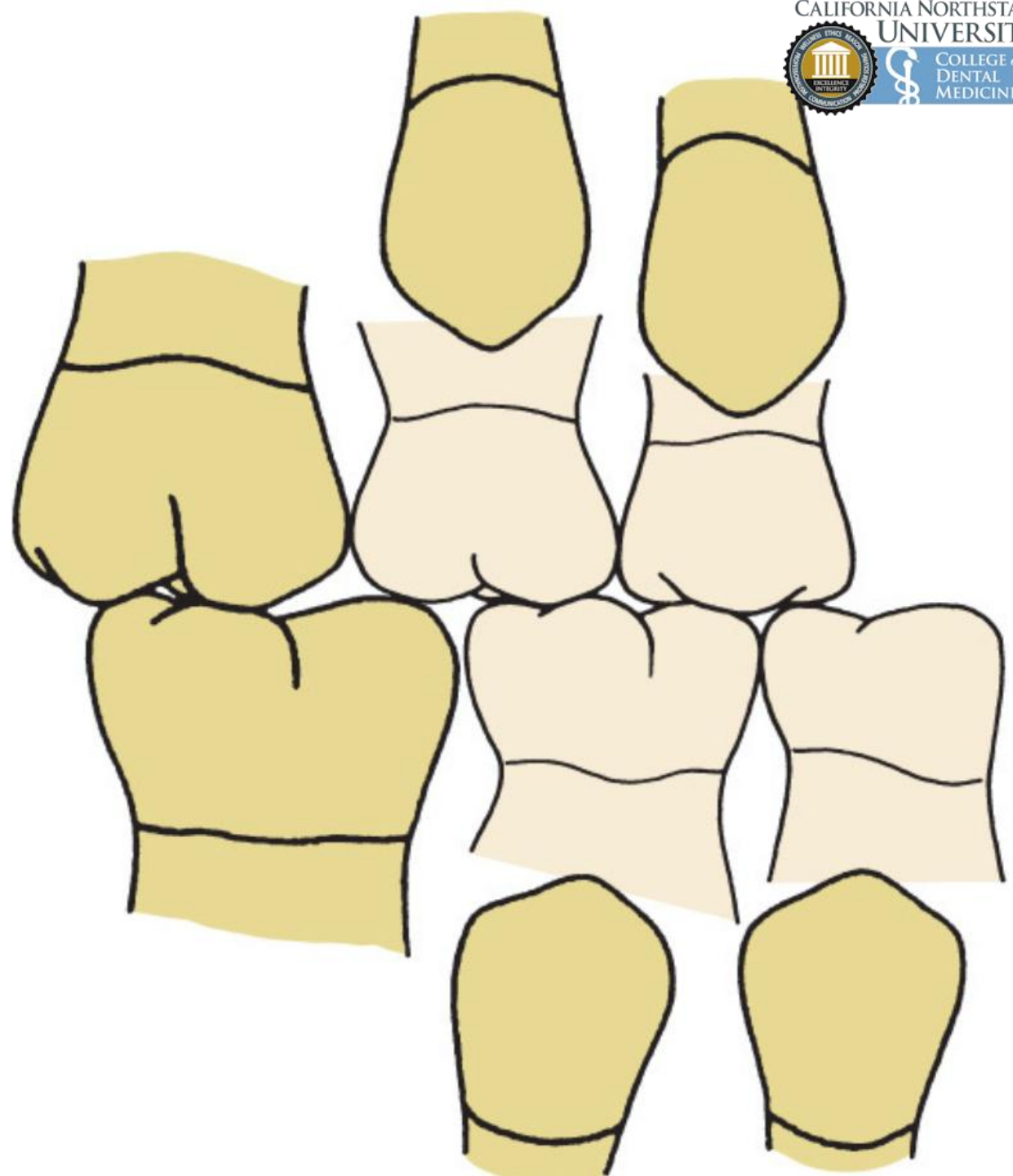
Primate Space

- Located mesial to the maxillary canine
- Located distal to the mandibular canine

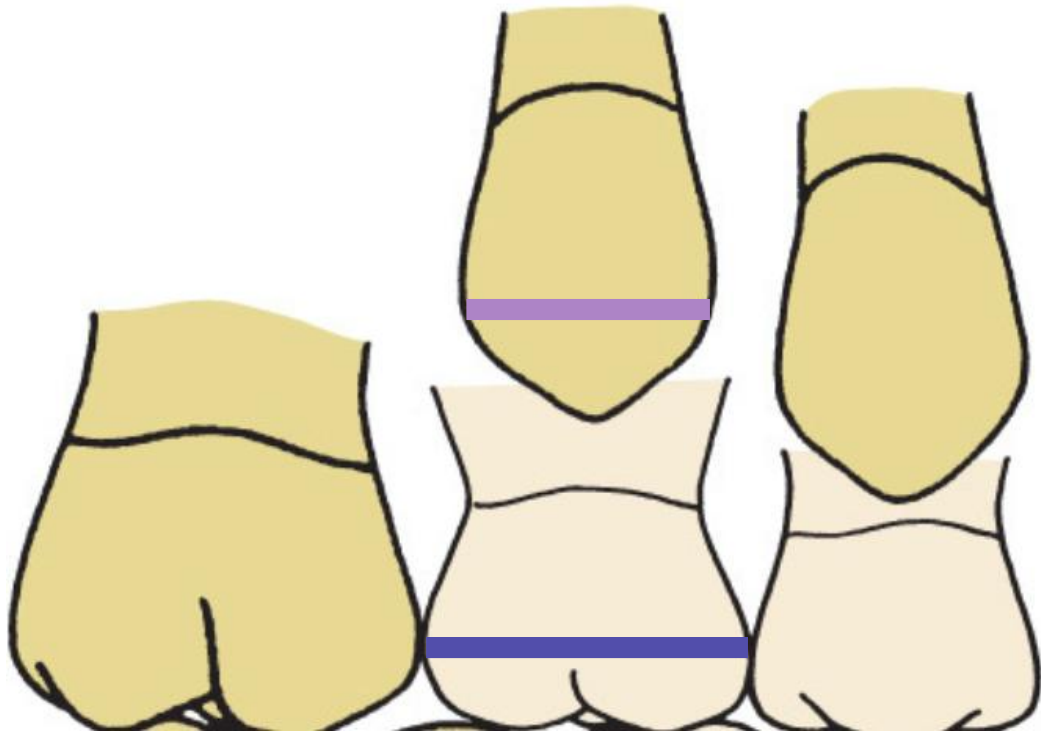


“E” Space

- The space difference between the second primary molar and the second permanent premolar
- ~2.0 mm in the mandible
- ~1.5 mm in the maxilla

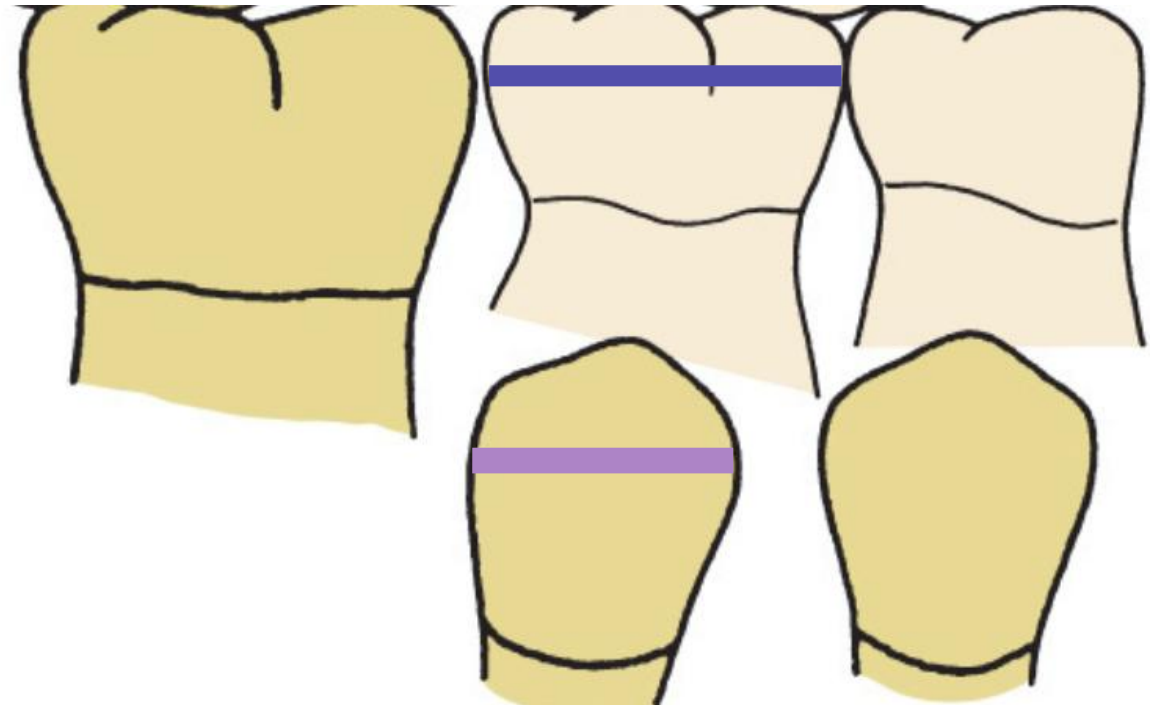


“E” Space



 = 1.5 mm

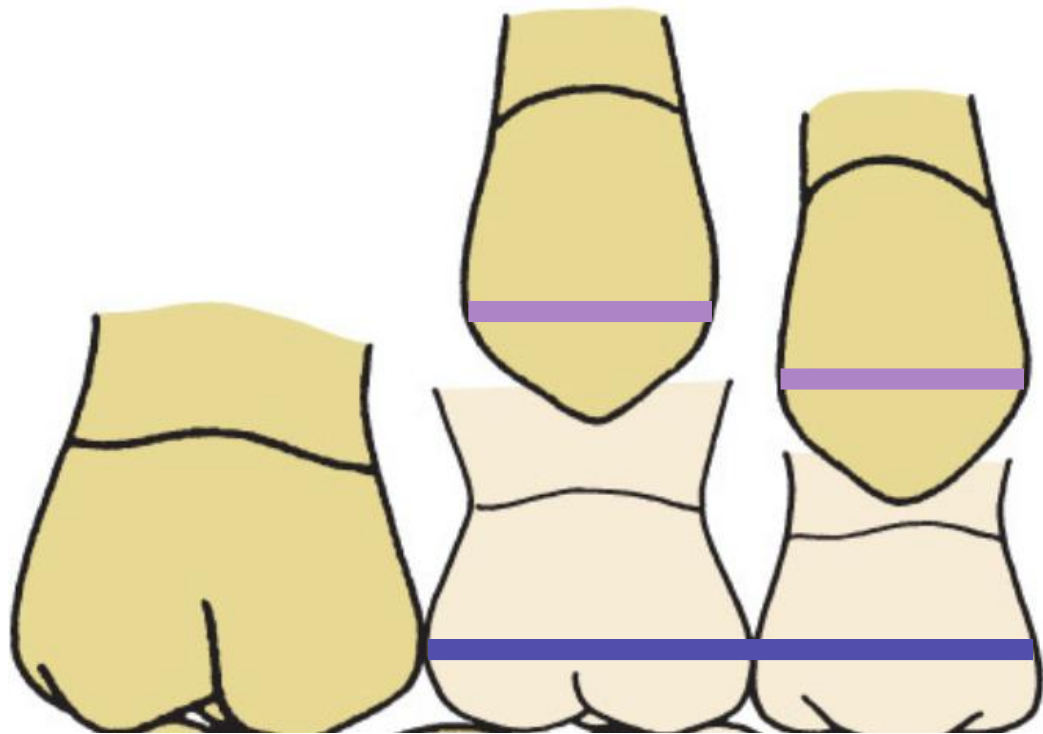

 = 2.0 mm

Leeway Space

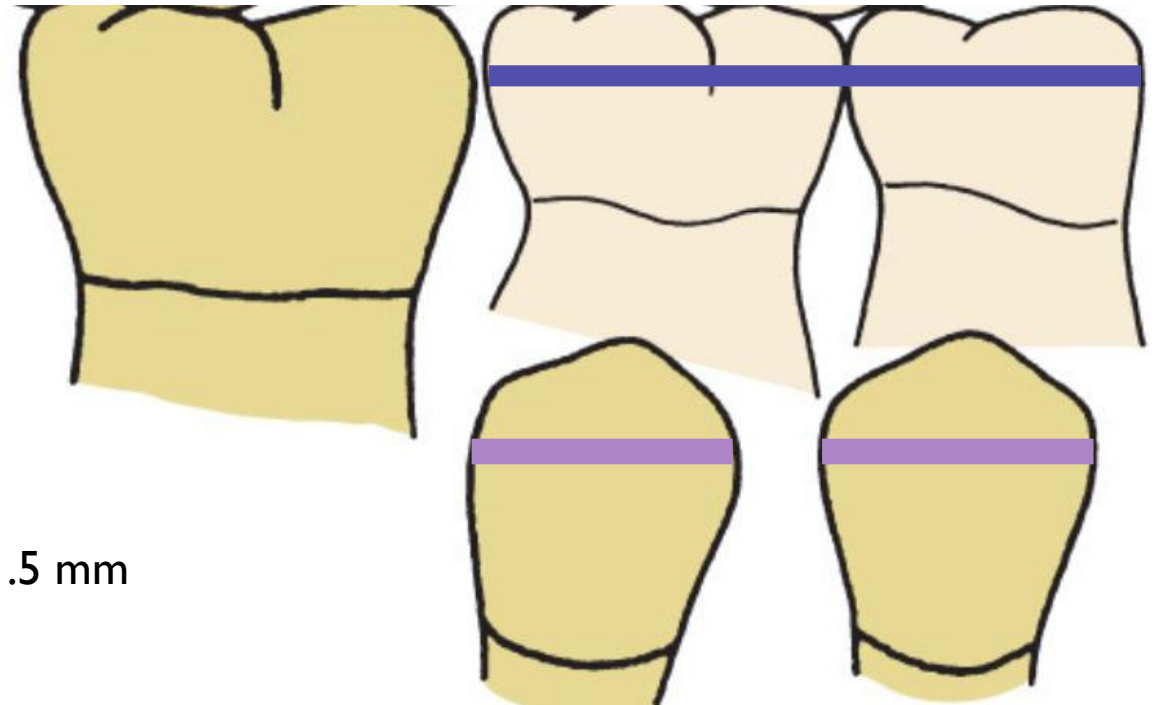
- Leeway space considers the size difference of both primary molars compared to both premolars
- In the maxilla the first primary molar and first permanent premolar are often similar in size
- In the mandible the first primary molar is about 0.5mm larger than the first permanent premolar
- In the maxilla the Leeway Space $\approx$ E Space $\approx$ 1.5mm
- In the mandible the Leeway Space $\approx$ E Space + 0.5mm $\approx$ 2.5 mm

Leeway Space



= 1.5 mm

= 2.5 mm



Variation

The Leeway and E space varies significantly based on

- Tooth size variation
- Eruption pattern

Leeway space can range from 0mm to 2.77 mm per side

Spaces tend to be

- Symmetric
- Larger in the mandible than the maxilla

Transition of the Dentition

Getting your big kid teeth

Why do we
care about
eruption?

Eruption is how teeth
assume their initial position
in the mouth

Understanding eruption can
help you identify issues and
concerns before they are
obvious

Phases

Primary Dentition Eruption (0-2 years)

Primary Dentition (2-6 years)

Early Mixed Dentition (6-9 years)

Late Mixed Dentition (9-12 years)

Permanent Dentition (12+ years)

Call Back!

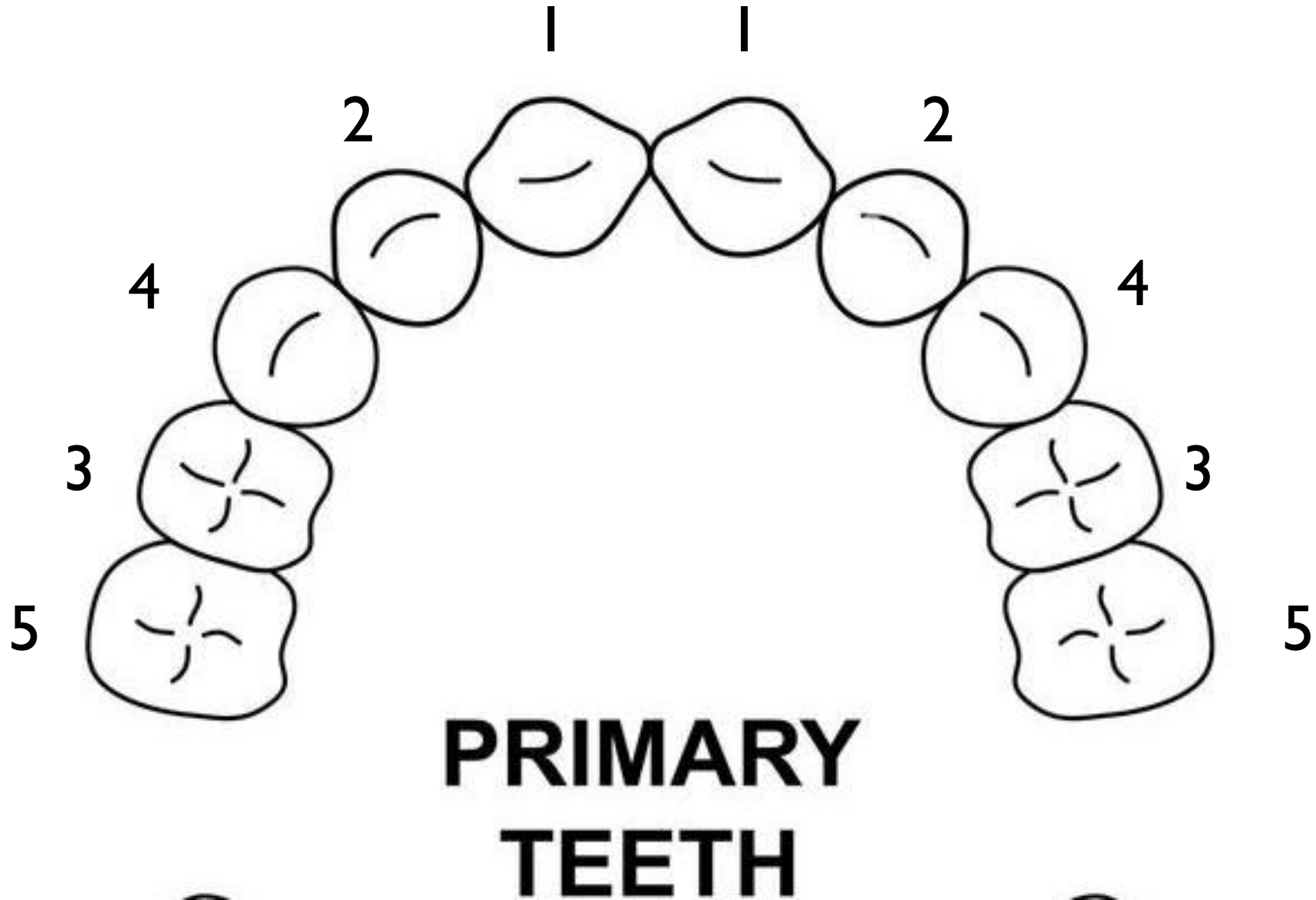
Primary Dentition

	Calcification begins at	Formation complete at	Eruption		Exfoliation	
			Maxillary	Mandibular	Maxillary	Mandibular
Central incisors	4 th fetal mo	18-24 mo	6-10 mo	5-8 mo	7-8 y	6-7 y
Lateral incisors	4 th fetal mo	18-24 mo	8-12 mo	7-10 mo	8-9 y	7-8 y
Canines	4 th fetal mo	30-39 mo	16-20 mo	16-20 mo	11-12 y	9-11 y
First molars	4 th fetal mo	24-30 mo	11-18 mo	11-18 mo	9-11 y	10-12 y
Second molars	4 th fetal mo	36 mo	20-30 mo	20-30 mo	9-12 y	11-13 y



Primary Tooth Eruption

- 20 primary teeth erupt by around 24 months of age
- After primary tooth eruption things largely stay the same until about 6 years of age
- The mandible tends to be earlier than the maxilla
- The first molar tends to erupts prior to the canine in both arches



On Eruption

There is huge variation in human physiology

Eruption is a process that is still not well understood

Many things may speed up or slow down eruption

Sequence is more important than exact timing

Exfoliation

The process of losing a primary/deciduous tooth is called exfoliation

Generally – exfoliation precedes eruption of a “succedaneous tooth”

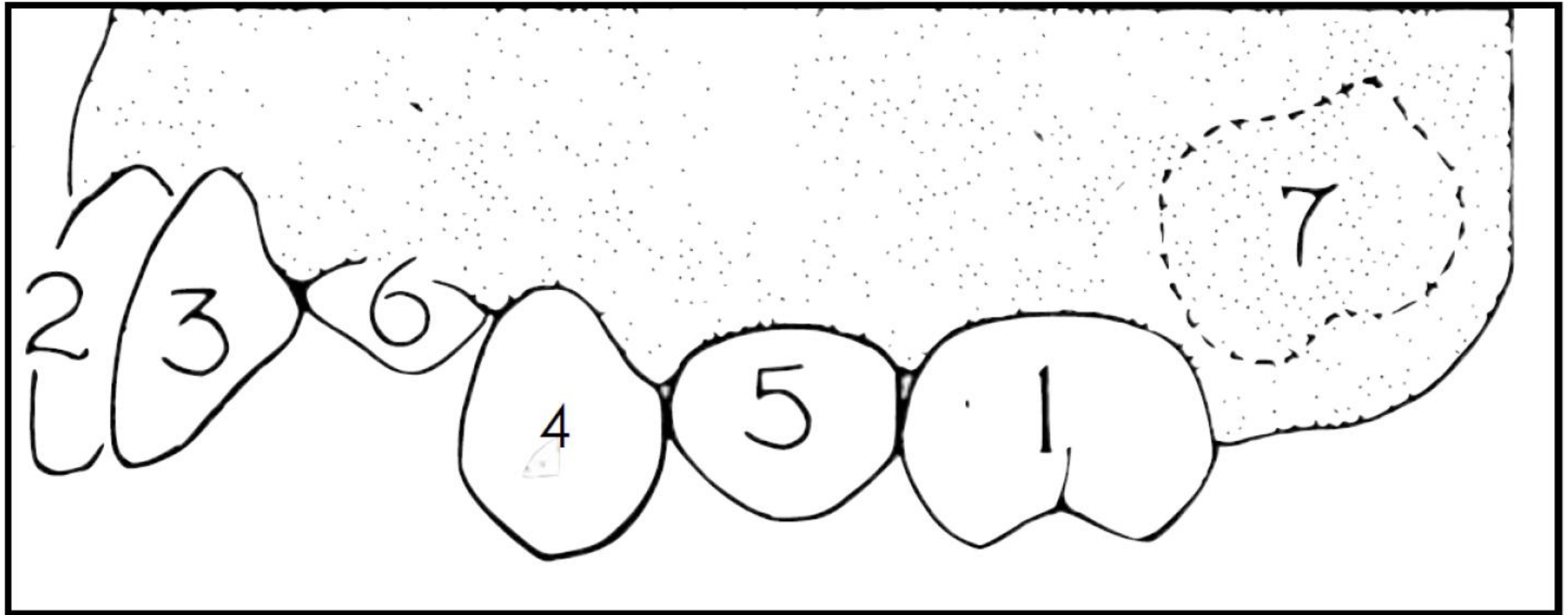
Exfoliation occurs through resorption of the cementum and dentin to the approximately the CEJ



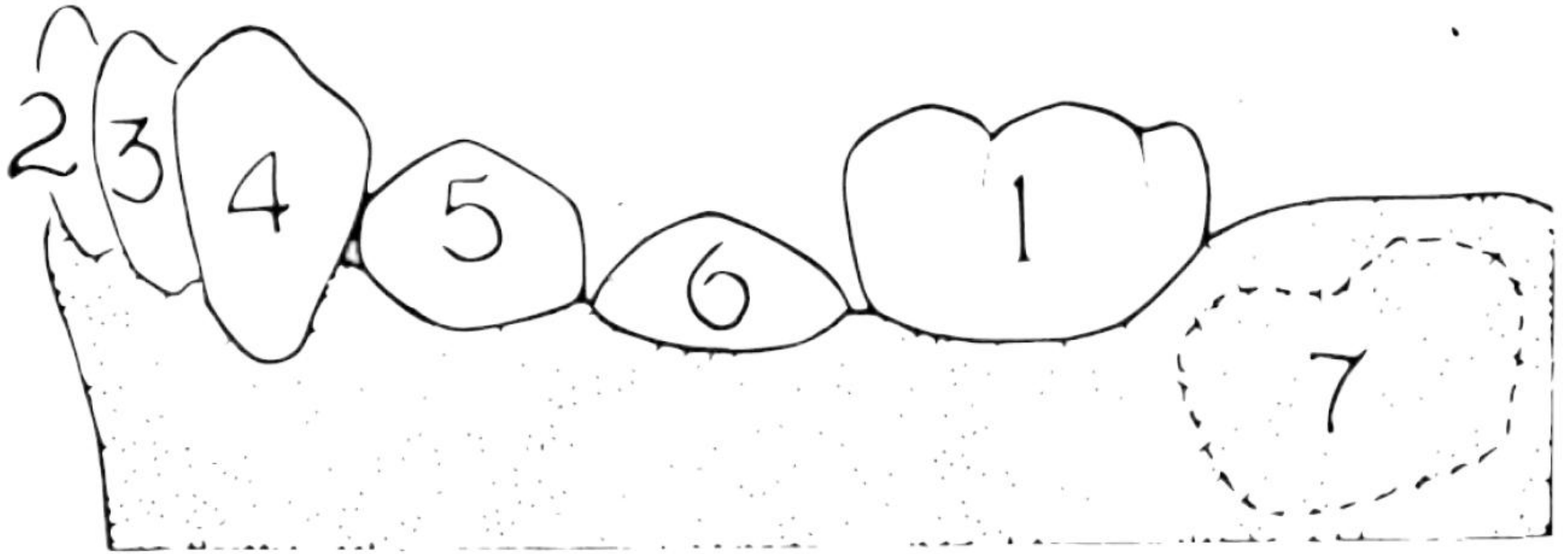
Call Back!

Permanent Dentition					
	Calcification begins at	Crown (enamel) complete at	Roots complete at	Eruption*	
				Maxillary	Mandibular
Central incisors	3-4 mo	4-5 y	9-10 y	7-8 y (3)	6-7 y (2)
Lateral incisors	Maxilla: 10-12 mo Mandible: 3-4 mo	4-5 y 4-5 y	11 y 10 y	8-9 y (5)	7-8 y (4)
Canines	4-5 mo	6-7 y	12-15 y	11-12 y (11)	9-11 y (6)
First premolars	18-24 mo	5-6 y	12-13 y	10-11 y (7)	10-12 y (8)
Second premolars	24-30 mo	6-7 y	12-14 y	10-12 y (9)	11-13 y (10)
First molars	Birth	30-36 mo	9-10 y	5.5-7 y (1)	5.5-7 y (1a)
Second molars	30-36 mo	7-8 y	14-16 y	12-14 y (12)	12-14 y (12a)
Third molars	Maxilla: 7-9 y Mandible: 8-10 y			17-30 y (13)	17-30 y (13a)

Maxillary



Mandibular



N. J.

Another Note

Ages of eruption are an average

There is huge variation in the actual eruption time for each individual

Pay attention to sequence and position over timing

If someone is late – they tend to be consistently late

If someone is early – they tend to be consistently early

Early Mixed Dentition

Exfoliation of

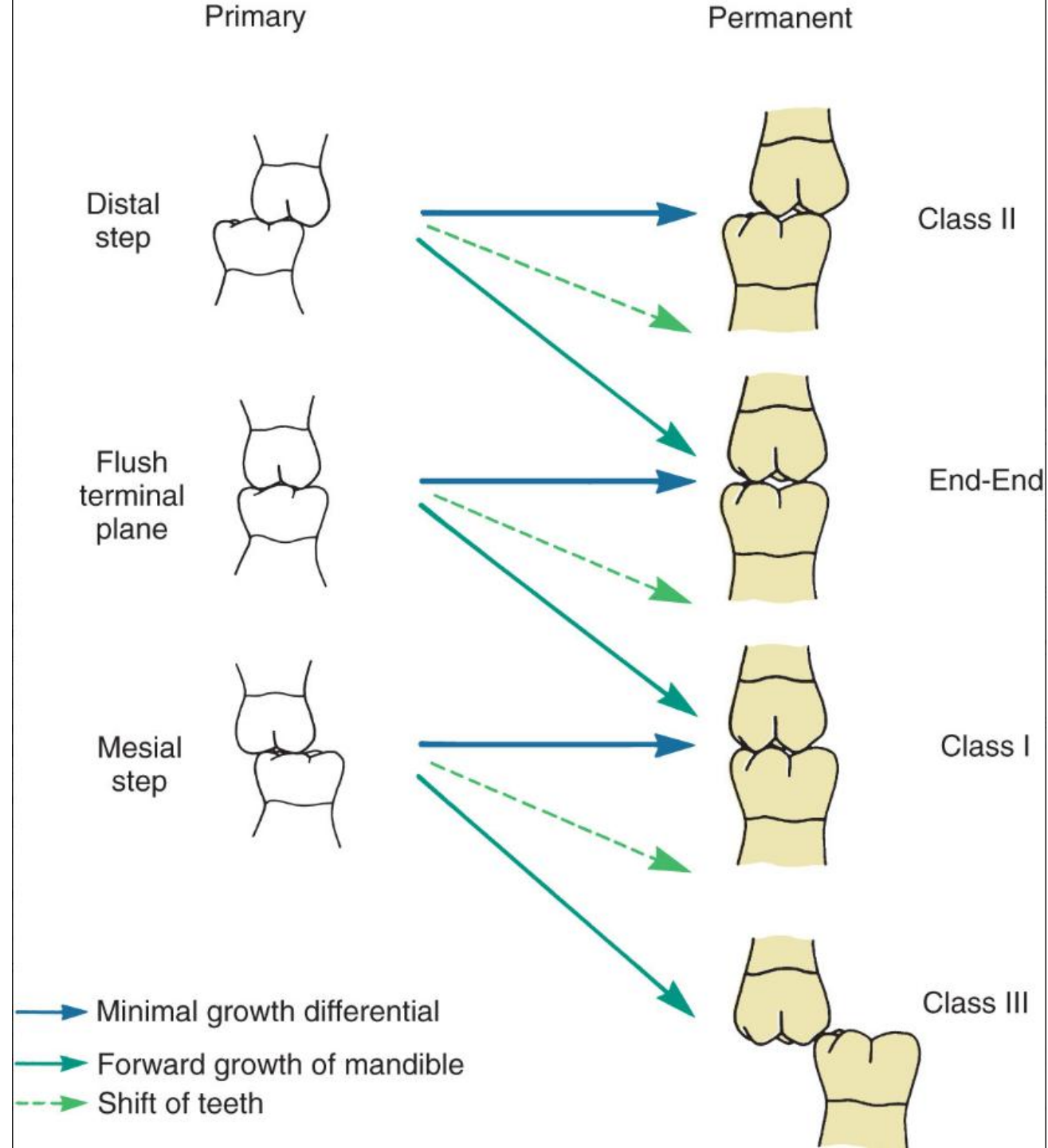
- Primary Centrals
- Primary Laterals

Eruption of the

- First Permanent Molars
- Permanent Central Incisors
- Permanent Lateral Incisors

Incisor Liability

Early Mesial Shift



Incisor Liability

The permanent incisors are larger than the primary incisors

This difference is termed “Incisor Liability”

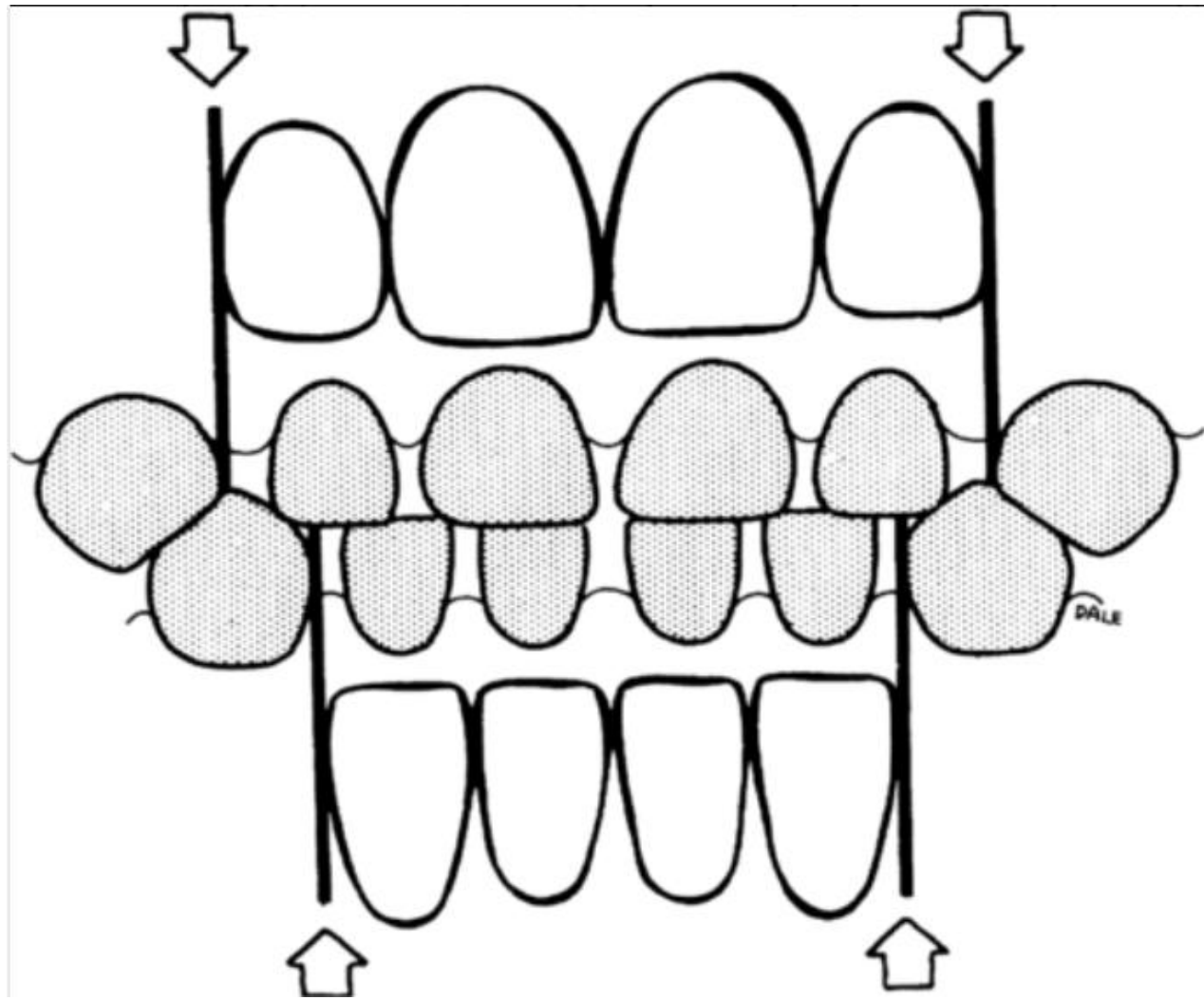
~7mm in the maxilla

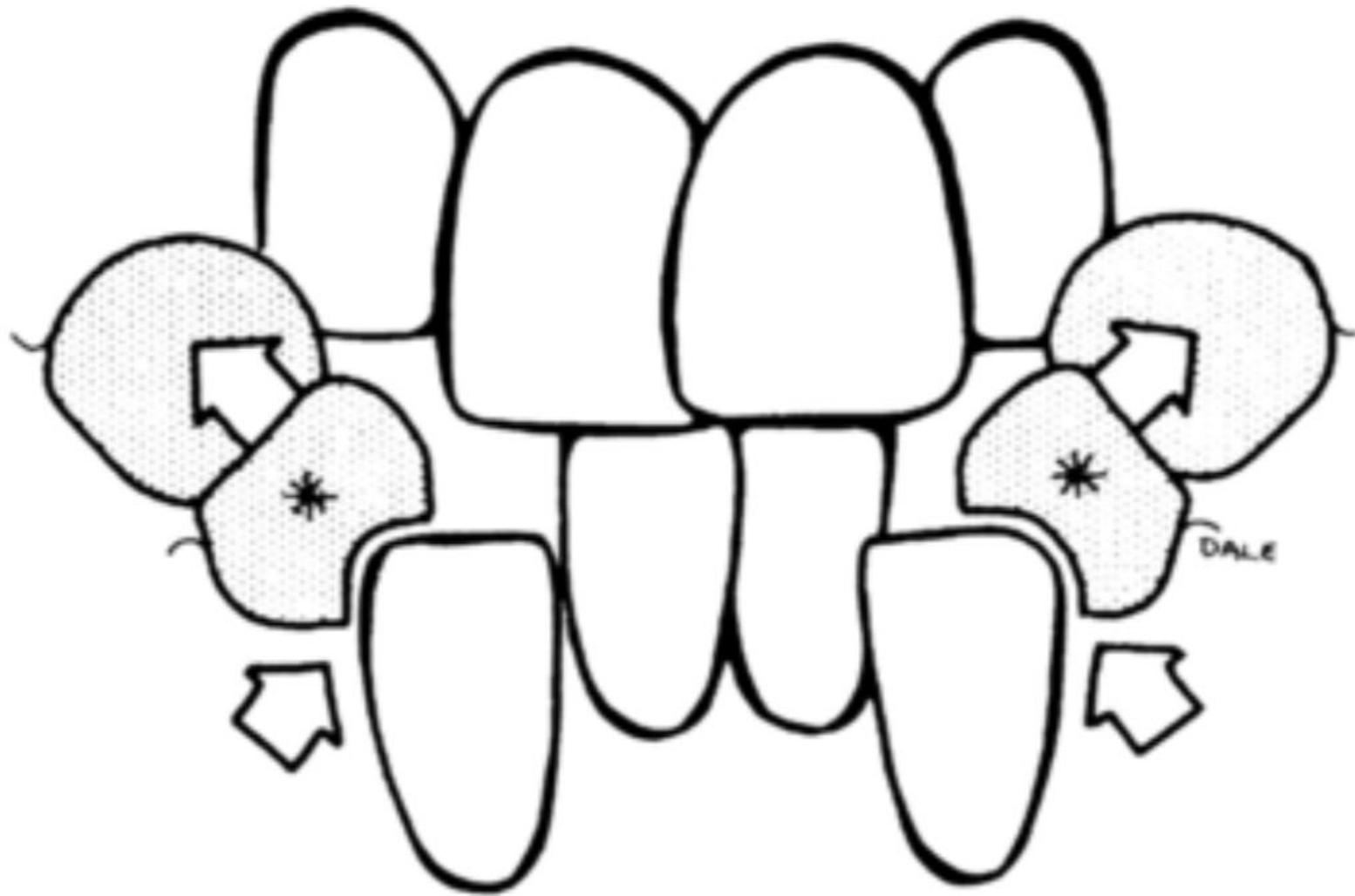
~5mm in the mandible

Incisor Eruption

Well spaced primary dentition can accommodate incisor liability

The less space available the more likely that the permanent teeth erupt in a crowded position





Spacing

Spacing in the primary dentition
correlates to permanent tooth position

Spacing 3-6 mm = No crowding

Spacing $< 3\text{mm}$ = 20% incisor crowding

No spacing = 50% incisor crowding

Crowded Primary teeth = 100% incisor
crowding

Early Mesial Shift

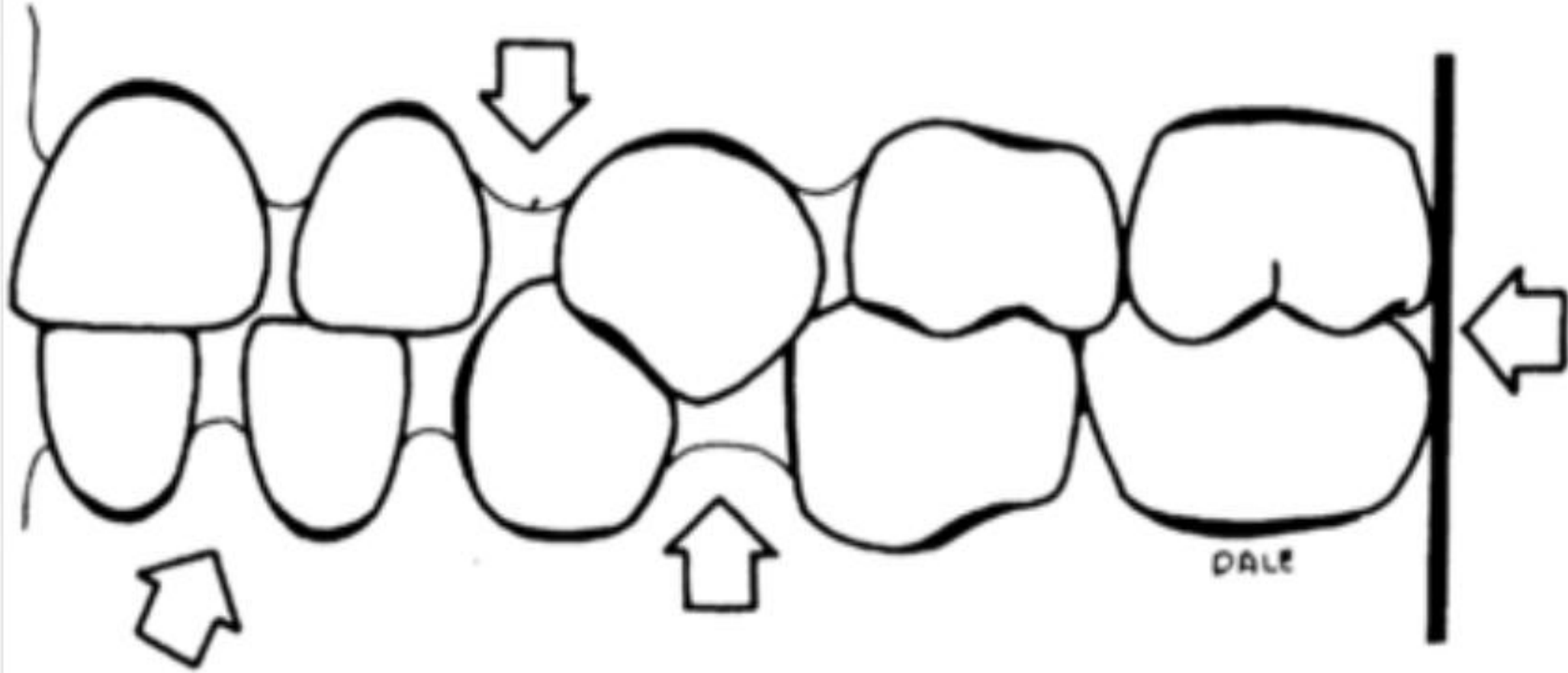
Eruption of the permanent first molars (3, 14, 19 and 30) results in a mesial motion forward of the posterior primary teeth

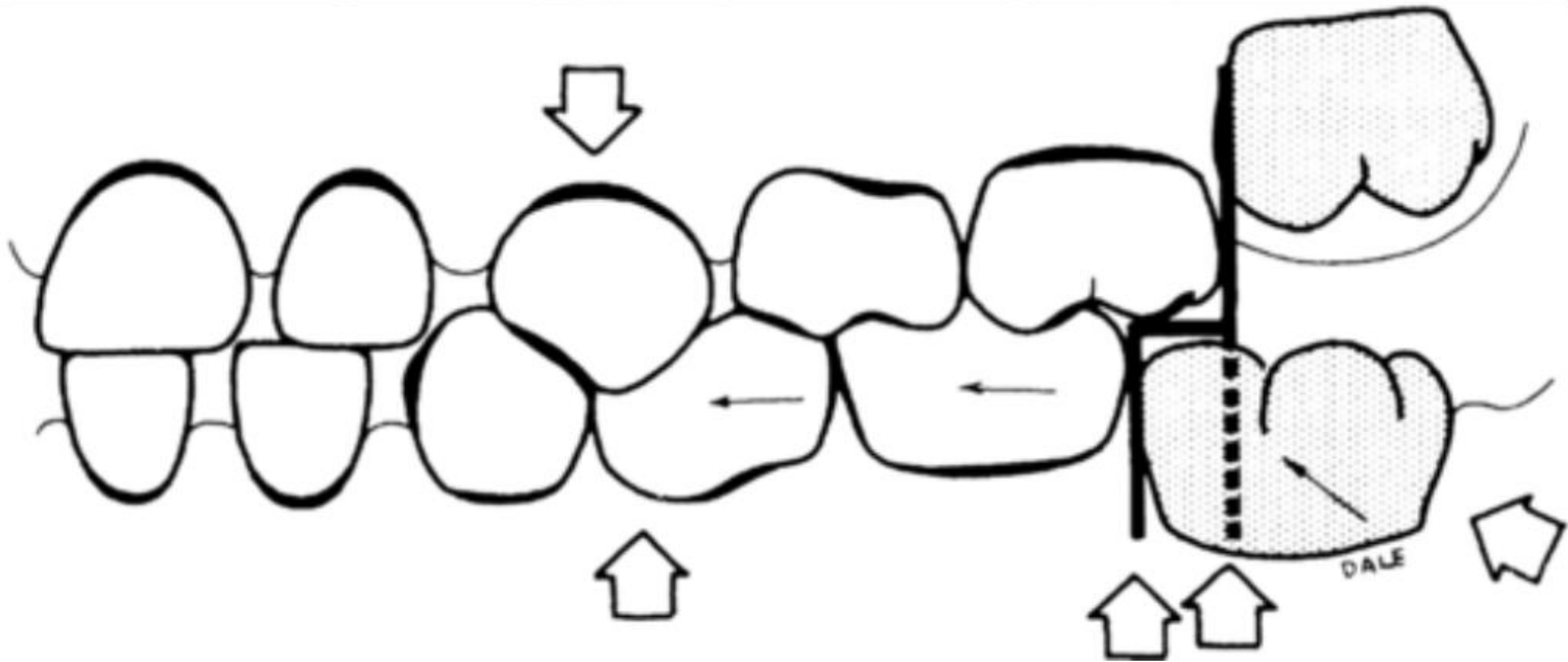


This motion reduces or closes

Posterior
interdental space

Primate spaces





Late Mixed Dentition

Exfoliation of

- Primary Canines
- Primary First Molar
- Primary Second Molar

Eruption of the

- Permanent Canines
- Permanent 1st Premolar
- Permanent 2nd Premolar
- Permanent 2nd Molar

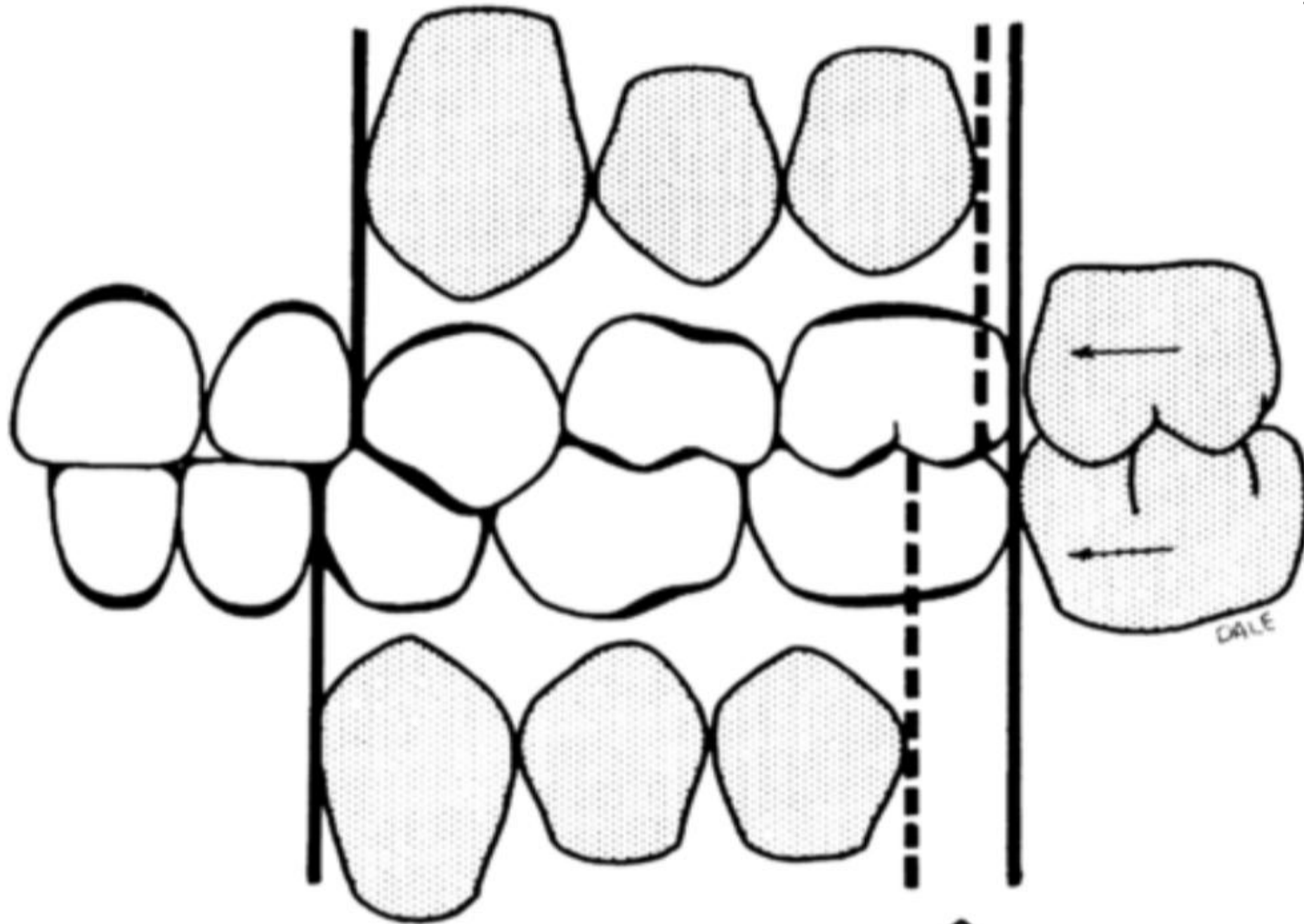
Late Mesial Shift Occurs

Late Mesial Shift

As the primary molars and canines are lost teeth continue to shift forward into the Leeway/E space

This closes existing spaces – which is often desirable

With adequate leeway space incisor crowding can be relieved by distal movement into this space



Space Closure Gone Wrong

Space Loss

Concerns with Spacing

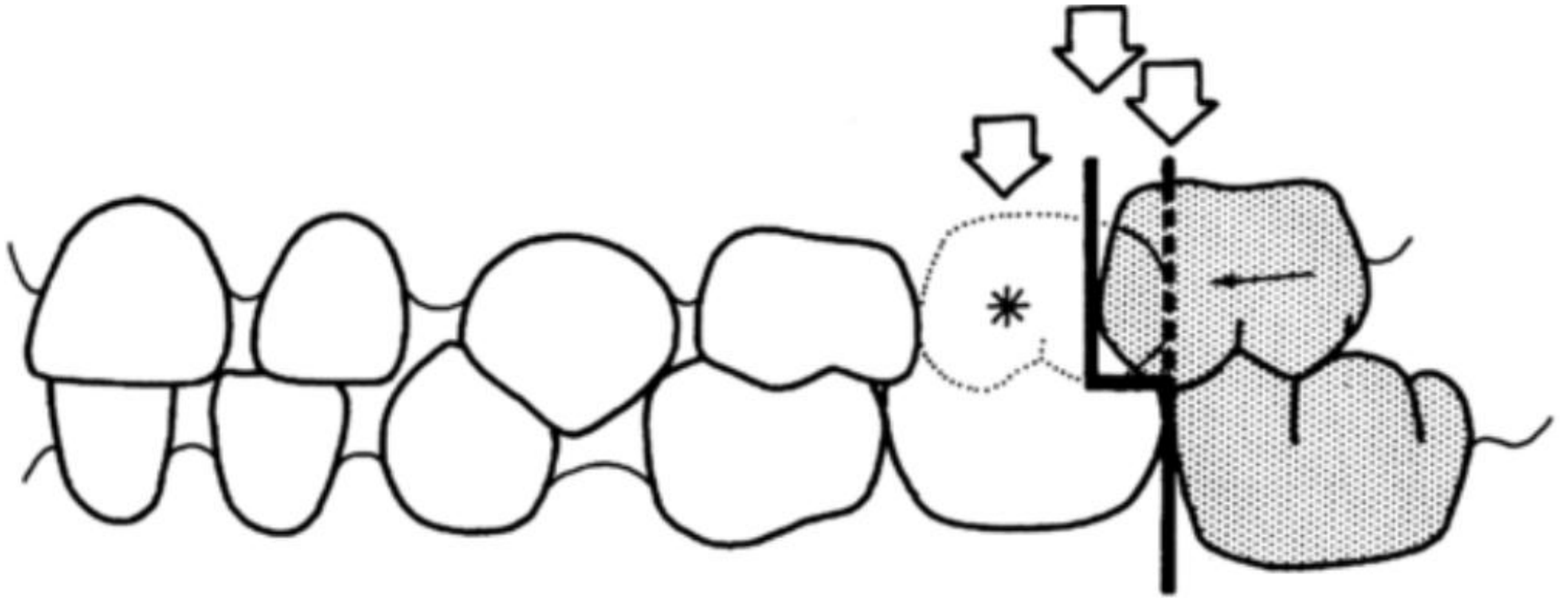
- Adequate spacing is essential for proper eruption of the dentition
- Teeth tend to move/tip mesially if there is nothing preventing it
- Space closure is a feature of the early and late mesial shifts
- Premature loss of teeth can result in several problems in the dentition

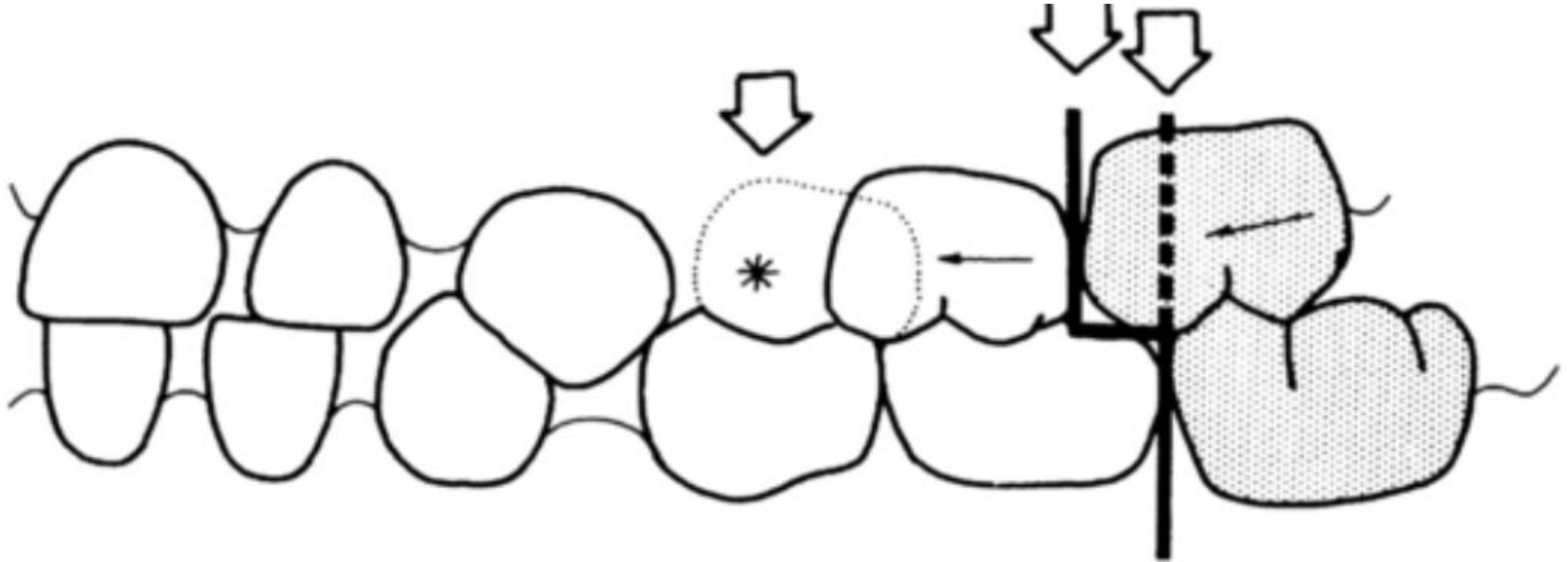
Premature Tooth Loss

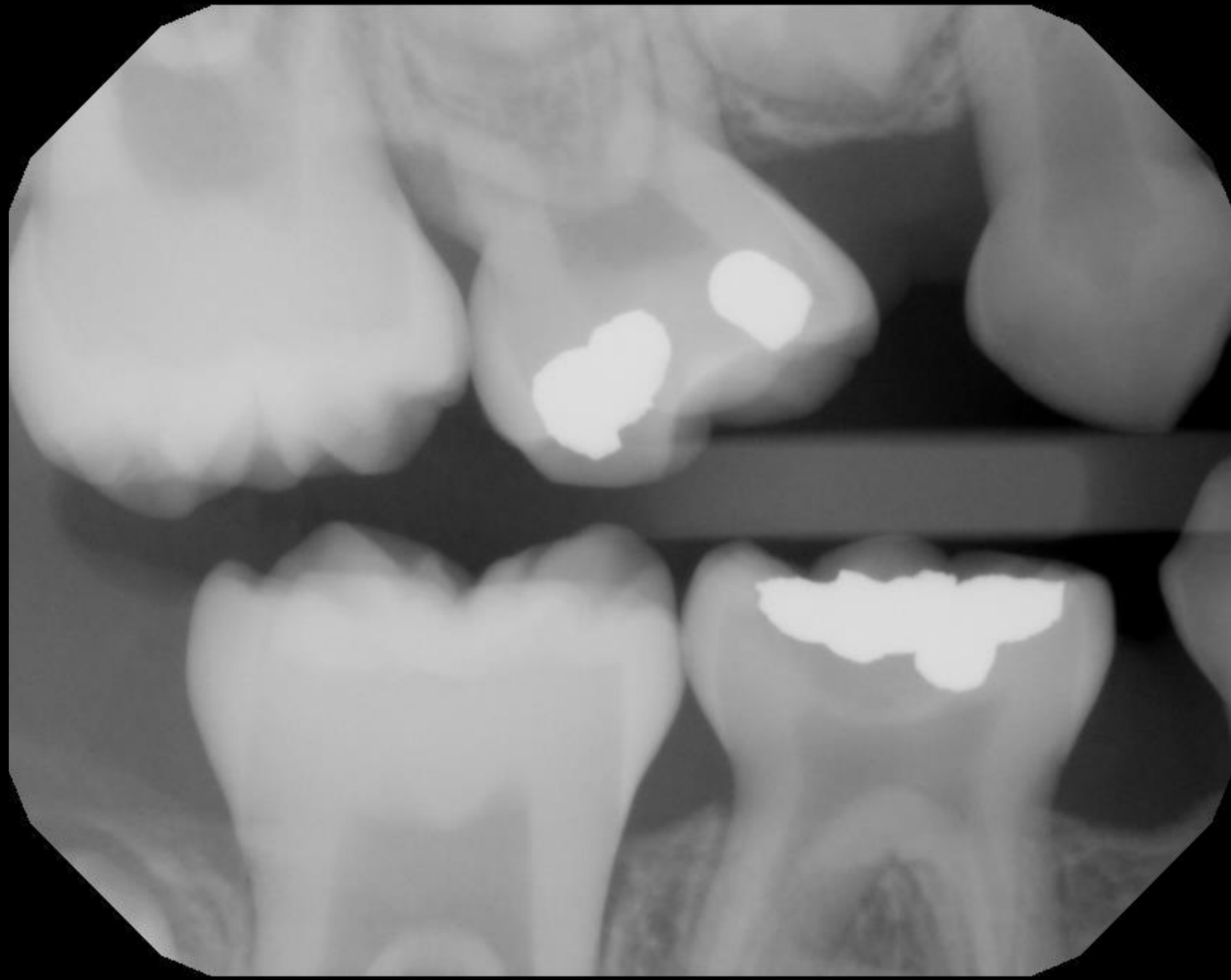


Features of Space Loss

- Early loss of posterior primary teeth almost always causes a reduction of available space
- Spaceloss occurs rapidly in the first 6 months following tooth loss then slowly afterwards
- Eruption of surrounding teeth accelerates space loss regardless of time since loss
- Maxillary teeth tend to move mesially and rotate in a ML direction
- Mandibular teeth tend to tip mesially with limited movement







Premature loss of anterior teeth

Less concerning for space loss as
teeth tend to shift less

Incisors

- Early incisor loss is usually the least detrimental, but impacts aesthetics the most
- May affect speech skills if not yet developed

Canines

- Early loss may shift midlines to **affected** side

Space Loss Concerns

Crowding

Prevention of eruption

Supraeruption of opposing arch

More likely to need orthodontic
intervention

Difficulty with speech and mastication

Prevention of Space Loss

Maintain the
dentition

Provide space
maintenance

Space Maintenance

Anything that prevents movement of teeth in an undesired direction

If there is something in the way — teeth don't move

Ideal design minimizes physiologic impact and maximizes retention

Space Maintenance Selection

Identify key characteristics

- Tooth/teeth lost
- Available anchors
- Available abutments

Decision to proceed

- Stage of dental development
- Patient Factors

Space Maintainer Indications

Early Loss of Tooth

Concern for space management

Upcoming active eruption

Upcoming mesial shifts

Pediatric
Partial
(Groper
fixed
bridge)

Tooth/teeth lost: Anterior
Tooth

Available anchors: Stable
Molar Teeth

Available abutments:
Edentulous anterior ridge



Pediatric Partial (Groper fixed bridge)

Bands placed on most stable molars

Wire loop with acrylic pad and teeth extends anteriorly

Fills in missing dentition

Primarily an esthetic replacement – not spaceloss prevention

Band and Loop

Tooth/teeth lost: Single Tooth

Available anchors: Tooth on the mesial or distal of lost tooth

Available abutments: Tooth on opposite side of edentulous space as anchor



C



D

Band and Loop

A band is placed on an anchorage tooth

A wire loop extends forward to an abutment tooth

Because of limited strength of the loop only a single tooth space can be preserved with a band and loop appliance

Nance Appliance

Tooth/teeth lost: One or more *maxillary* teeth

Available anchors: Teeth distal to the edentulous spaces

Available abutments: Healthy hard palate



Nance Appliance

Bands are placed on distal most teeth

A wire loop connects to an acrylic pad

The pad contacts the hard palate

Requires regular removal and cleaning

Lower Lingual Holding Arch

Tooth/teeth lost: One or more
mandibular teeth

Available anchors: Teeth distal
to the edentulous spaces

Available abutments:
Permanent mandibular incisors



B

LLHA

Bands are placed on the distal most teeth

A wire curves forward to contact the lingual of the lower **permanent** incisors

Must be careful to avoid impinging on soft tissue

Distal Shoe

Tooth/teeth lost: One
primary second molar

Available anchors:
Primary first molar

Available abutments:
None



C





Distal Shoe

Band is placed on the primary first molar

Wire loop extends distally and connects to a plane

The plane enters the soft tissue and is positioned as close as possible to the unerupted permanent first molar

This requires exposure of the molar either surgically or during extraction of the primary tooth

Must be monitored and replaced as soon as feasible



Key features for anchorage/abut ment

Permanent tooth anchorage is always preferred

Bands should fit well with minimal gingival pressure

Abutment contact should be just below the height of contour on teeth

Healthy teeth preferred for abutments

Space Maintainer Tips

- Place as soon as possible after tooth loss
- Always account for occlusion to avoid interference
- Bilateral spacers avoid blocking permanent tooth eruption and are generally preferred
- Spacers require periodic removal and cleaning
- Spacers should be removed ~6 months from expected permanent tooth eruption

Space Maintainer Contraindications

Poor hygiene

Near eruption

Long time since tooth loss

Unfavorable abutment/anchorage

Immune status for distal shoes

Early Orthodontic Problems

Interceptive Concerns

Recognizing Early Ortho Problems

Identify problems that are

- Lasting
- Progressive
- Easier to treat early

Reassure parents about problems that

- Self-resolve
- Stay stable
- Are best treated as part of
comprehensive care

Lasting vs. Self Resolving Problems

Many things change and are not ideal in the transition from primary to permanent dentition

Not all of them require intervention, since many will improve or resolve naturally

Some occlusal concerns are unlikely to resolve on their own

Progressive vs. Stable

Some problems are present, but will stay the same through growth

Others will worsen with growth – these often benefit from intervention

Easier to
treat early

Younger individuals have
less dense bone

Dental eruption and skeletal
growth can be manipulated
to more favorable patterns
in some cases

Things that
are not so
concerning

The ugly duckling phase

Spacing

Lingual Eruption

Mild crowding









Things that
are more
concerning

Class II or III
skeletal
relationships

Crossbites

Severe
Crowding

Severe
malposition

Ectopic
eruption

Impaction

















Correction Mechanisms

Fixing Bites

Interventions

RPE

Bite
turbos

Headgear

Herbst
Appliance

Halterman
Appliance



Rapid Palatal Expander (RPE)

*Correct posterior
crossbite*

Gain transverse arch
width

Create space for other
movement



Bite Turbos

Prop open teeth for correct
of anterior crossbite

Prevent opposing arch from
interfering with movement

Very often in conjunction
with other appliances

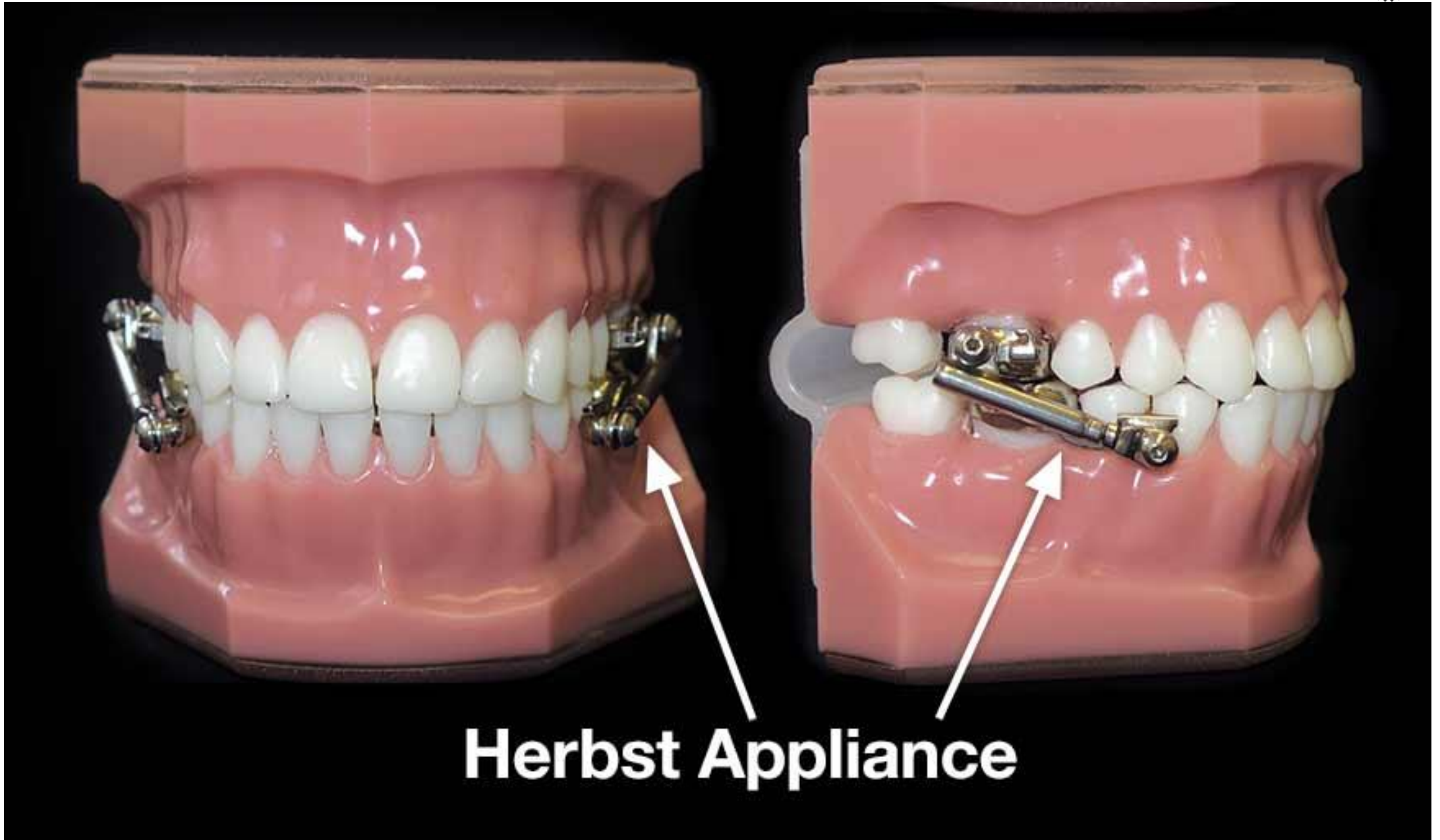


Headgear

Applies anterior force on the maxilla to help correct Class III caused by maxillary deficiency

Can be modified to suit other situations

Utilizes and/or prevents natural growth patterns



Herbst Appliance

Advances the mandible to encourage anterior growth

Helps correct class II from a deficient mandible

Utilizes natural growth patterns



Halterman Appliance

Distalizes erupting teeth by
using elastics

May require surgical exposure
of the erupting tooth

Multiple Designs possible

Orthodontic Appliances

There are many, many more options available

Remember at its heart orthodontics is all about forces and biomechanics

If you push on something long enough, it moves

Questions?
